

Business Problem Statement

Retail Consumer Shopping Behavior Analysis

Background

A leading retail company wants to gain a deeper understanding of how its customers shop, with the goal of increasing sales, improving customer satisfaction, and building long-term loyalty. Management has observed shifts in purchasing patterns across customer demographics, product categories, and sales channels (online versus in-store).

The company is especially keen to identify which factors influence consumer decisions and drive repeat purchases, including discounts, customer reviews, seasonality, and preferred payment methods.

Objective

Using the company's consumer behavior dataset, this project aims to answer the following overarching business question:

“How can the company leverage consumer shopping data to identify trends, strengthen customer engagement, and optimize its marketing and product strategies?”

Deliverables

Deliverable	Tool	Description
1. Data Preparation & Modeling	Python	Clean, validate, and transform the raw dataset into an analysis-ready format.
2. Data Analysis	SQL	Organize the data into a structured relational schema, simulate business transactions, and run queries to uncover insights on customer segments, loyalty, and key purchase drivers.
3. Visualization & Insights	Power BI	Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
4. Report & Presentation	Documentation	Write a clear project report summarizing key findings and business recommendations, and prepare a presentation that communicates insights and actionable next steps to stakeholders.
5. GitHub Repository	Version Control	Publish all Python scripts, SQL queries, and dashboard files in a well-organized, documented repository.